53.6: The human population is no longer growing exponentially but is still increasing extremely rapidly

In the last few centuries, the human population has grown at an unprecedented rate, more like the elephant population in Kruger National Park (see Figure 53.8) than the fluctuating populations we considered in Concept 53.5. No population can grow indefinitely, however. In this section of the chapter, we'll apply the concepts of population dynamics to the specific case of the human population.

The Global Human Population

The human population has grown explosively over the last four centuries **(Figure 53.21)**. In 1650, about 500 million people inhabited Earth. Our population doubled to 1billion within the next two centuries, doubled again to 2billion by 1930, and doubled still again to 4billion by 1975. Notice that the time it took our population to double in size decreased from 200 years in 1650 to just 45 years in 1930. Thus, historically our population has grown even *faster* than exponential growth, which has a constant rate of increase and hence a constant doubling time.

Figure 53.21

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Human population growth (data as of 2022).
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A graph plots Human population in billion versus Year from 8000 B C E to the present day. The curve rises slightly up until approximately 1800 when it exceeds 1 billion. It then rises sharply to 7.5 for the present day.

The global human population has grown almost continuously throughout history, but it skyrocketed after the Industrial Revolution.

The global population is now more than 7.9 billion people and is increasing by about 75 million each year. This translates into more than 200,000 people each day, the equivalent of adding a city the size of Amarillo, Texas. At this rate, it takes only about four years to add the equivalent of another United States to the world population. Ecologists predict there will be about 9.7 billion people on Earth by the year 2050.

Though the global population is still growing, the *rate* of growth began to slow during the 1960s **(Figure 53.22)**. The annual rate of increase in the global population peaked at 2.2% in 1962 but was only 0.9% in 2022. Current models project a growth rate of 0.5% by 2050, which would add about 50million more people per year if the

population climbs to a projected 9.7 billion. The reduction in annual growth rate observed over the past six decades shows that the human population is now growing more slowly than expected in exponential growth. This change resulted from fundamental shifts in population dynamics due to diseases, including AIDS, and to voluntary population control.

Figure 53.22

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Annual percent increase and recent human population growth.
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A graph plots annual percent increase and human population versus year from 1950 to 2050.

The annual percentage increase rises and falls between 1950 and 1975, staying between 1.3 and 2.2. It then steadily falls to 0.5 in 2050. Human population rises from 2.5 in 1950 to 9.7 in 2050. The graphs intersect near 2002. Data after the present day are projected.

The sharp dip in the rate of increase around 1960 is due mainly to a famine in China in which about 60 million people died. (Data as of 2022.)

VISUAL SKILLS *The blue curve in this figure appears to tell a different story about human population growth than does the curve in*Figure 53.21*, yet both curves are accurate. Summarize what each curve conveys about recent human population growth and explain how both can be accurate.*

*For suggested answer, see*Appendix A*.*

Regional Patterns of Population Change

We have described changes in the global population, but population dynamics vary widely from region to region. In a stable regional population, birth rate equals death rate (disregarding the effects of immigration and emigration). Two possible configurations for a stable population are

Zero population growth=High birth rate−High death rateorZero population growth=Low birth rate−Low death rate

The movement from high birth and death rates toward low birth and death rates, which tends to accompany industrialization and improved living conditions, is called the **demographic transition**. In Sweden, this transition took about 150 years, from 1810 to 1975, when birth rates finally approached death rates. In Mexico, where the human population is still growing rapidly, the transition is projected to take until at least 2050. Demographic transition is associated with an increase in the quality of health care and sanitation as well as improved access to education, especially for women.

After 1950, death rates declined rapidly in most developing countries, but birth rates have declined more variably. Birth rates have fallen most dramatically in China. The expected number of children per woman per lifetime in China decreased from 5.9 in 1970 to 1.6 in 2011, largely due to the government's strict "one-child" policy. In some

countries of Africa, the transition to lower birth rates has also been rapid, though birth rates remain high in most of sub-Saharan Africa.

How do such variable birth rates affect the growth of the world's population? In industrialized nations, populations are near equilibrium, with reproductive rates near the replacement level of 2.1 children per female (over their lifetime). In many industrialized countries—including the United States, Canada, Germany, Japan, and the United Kingdom—total reproductive rates are in fact *below* the replacement level. These populations will eventually decline if there is no immigration and if the birth rate does not change. In fact, the population is already declining in many eastern and central European countries. Most of the current global population growth occurs in less industrialized countries, where about 80% of the world's people now live.

A unique feature of human population growth is our ability to control family sizes through planning and voluntary contraception. Social change and the rising educational and career aspirations of women in many cultures encourage women to delay marriage and postpone reproduction. Delayed reproduction helps to decrease population growth rates and to move a society toward zero population growth under conditions of low birth rates and low death rates. However, there is a great deal of disagreement as to how much support should be provided for global family planning efforts.

Age Structure

Another important factor that can affect population growth is a country's **age structure**, the relative number of individuals of each age in the population. Age structure is commonly graphed as "pyramids" like those in **Figure 53.23**. For Zambia, the pyramid is bottom heavy, skewed toward young individuals who will grow up and perhaps sustain the explosive growth with their own reproduction. The age structure for the United States is relatively even until the older, postreproductive ages. Although the current total reproductive rate in the United States is about 1.8 children per woman—below the replacement rate—the population is projected to grow slowly through 2050 as a result of immigration. For Italy, the pyramid has a small base, indicating that individuals younger than reproductive age are relatively underrepresented in the population. This situation contributes to the projection of a future population decrease in Italy.

Figure 53.23

Age-structure pyramids for the human population of three countries.

Age Structure Pyramid for Afghanistan, United States, Italy. Values are estimated.

Data are as follows:

Sex	Age Group (Years)	Percent of Population: United Afghanistan	Percent of Population: Population: States Italy	of
Male	0 to 4	8	3.4	2.5
Female	0 to 4	8	3.2	2.1
Male	5 to 9	7	3.5	2.5
Female	5 to 9	7	3.2	2.1
Male	10 to 14	7.1	3.5	2.5
Female	10 to 14	7.1	3.2	2.1
Male	15 to 19	5.8	3.6	2.5
Female	15 to 19	6	3.3	2.1
Male	20 to 24	4	3.6	2.5
Female	20 to 24	4.2	3.5	2.5
Male	25 to 29	3.8	3.5	2.6
Female	25 to 29	4	3.5	3
Male	30 to 34	3.8	3.5	3.1
Female	30 to 34	3.8	3.5	3.6
Male	35 to 39	3	3.5	3.8
Female	35 to 39	3	3.5	4
Male	40 to 44	2.7	3.6	3.8
Female	40 to 44	2.7	3.6	4
Male	45 to 49	2.5	3.8	3.8
Female	45 to 49	2.8	3.8	3.9
Male	50 to 54	1.3	3.8	2.6
Female	50 to 54	1.3	3.8	3.6
Male	55 to 59	1	3.5	3.6
Female	55 to 59	1	3.5	3.6

Male	60 to 64	0.8	3.5	3.6
Female	60 to 64	0.9	3.5	3
Male	65 to 69	0.7	3	2.2
Female	65 to 69	0.7	3	2.2
Male	70 to 74	0.5	1.9	2.5
Female	70 to 74	0.5	2	2.5
Male	75 to 79	0.3	1.2	2.2
Female	75 to 79	0.25	1.5	1.8
Male	80 to 84	0.2	0.9	1.5
Female	80 to 84	0.1	1.4	1.5
Male	85 plus	Negligible	0.7	1.3
Female	85+	Negligible	1.5	1.5

The annual growth rate was approximately 2.8% in Zambia, 1.0% in the United States, and 0.0% in Italy. Note that Zambia's pyramid has a different scale for its x-axis from that of the other two pyramids. (Data as of 2010.)

Animation

Analyzing Age-Structure Pyramids

image

Play

image

Play

Age-structure diagrams not only predict a population's growth trends but also can illuminate social conditions. Based on the diagrams in Figure 53.23, we can predict that employment and education opportunities will continue to be a problem for Zambia in the foreseeable future. In the United States and Italy, a decreasing proportion of younger working-age people will soon be supporting an increasing population of retired "boomers." This demographic feature has made the future of Social Security and Medicare a major political issue in the United States. Understanding age structures is necessary to plan for the future.

Infant Mortality and Life Expectancy

Infant mortality, the number of infant deaths per 1,000 live births, and *life expectancy at birth*, the predicted average length of life at birth, vary widely in different countries. In 2015, for example, the infant mortality rate was 71 (7.1%) in Angola but only 2 (0.2%) in Japan. Life expectancy at birth was 59 years in Angola but 85 years in Japan. These differences reflect the quality of life faced by children at birth and influence the reproductive choices parents make. If infant mortality is high, then parents may choose to have more children to ensure that some reach adulthood.

Although global life expectancy has been increasing since about 1950, it has recently dropped in a number of regions, including countries of the former Soviet Union and in sub-Saharan Africa. In these regions, social upheaval, decaying infrastructure, and infectious diseases such as AIDS and tuberculosis are reducing life expectancy.

Global Carrying Capacity

No ecological question is more important than the future size of the human population. As noted earlier, population ecologists project a global population of about 9.7 billion people in 2050. That means that nearly 2 billion people will be added to the population in the next three decades because of the momentum of population growth. But just how many humans can the biosphere support? Will the world be overpopulated in 2050? Is it *already* overpopulated?

Estimates of Carrying Capacity

For over three centuries, scientists have attempted to estimate the human carrying capacity of Earth. The first known estimate, 13.4 billion people, was made in 1679 by Anton van Leeuwenhoek (a Dutch scientist who also discovered protists). Since then, estimates have varied from less than 1 billion to more than 1,000 billion 1trillion.

Carrying capacity is difficult to estimate, and scientists use different methods to produce their estimates. Some researchers use curves like that produced by the logistic equation (see Figure 53.9) to predict the future maximum of the human population. Others generalize from existing “maximum” population density and multiply this number by the area of habitable land. Still others base their estimates on a single limiting factor, such as food, and consider variables such as the amount of farmland, the average yield of crops, the prevalent diet—vegetarian or meat based—and the number of calories needed per person per day.

Limits on Human Population Size

A more comprehensive approach to estimating the carrying capacity of Earth is to recognize that humans have multiple constraints: We need food, water, fuel, building materials, and other resources, such as clothing and transportation. The **ecological footprint** concept summarizes the aggregate land and water area required by each person, city, or nation to produce all the resources it consumes and to absorb all the waste it generates.

What is a sustainable ecological footprint for the entire human population? One way to estimate this footprint is to add up all the ecologically productive land on the planet and divide by the size of the human population. Typically, this estimate is made using *global hectares*, where a global hectare (gha) represents a hectare of land or water with a productivity equal to the average of all biologically productive areas on Earth (1 hectare=2.47 acres). This calculation yields an allotment of 1.7 gha per person—the benchmark for comparing actual ecological footprints. Anyone who consumes resources that require more than 1.7 gha to produce is using an unsustainable share of Earth's resources, as is the case for the citizens of many countries **(Figure 53.24)**. For example, a typical ecological footprint for a person in the United States is 8 gha. Globally, the average footprint is 2.7 gha per person, more than a 50% overshoot of the sustainable use (1.7 gha per person) of Earth's resources.

Figure 53.24

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Per capita ecological footprint by country.
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A map of world with regions highlighted in accordance with their per capita ecological footprint.

Map of world with regions highlighted in accordance with their per capita ecological footprint in global hectares (gha). Per capita ecological footprint is described here in terms of three categories: greater than 6 gha, 3 to 6 gha, and 0 to 3 gha.

Greater than 6 gha (red areas in map): United States, Canada, Finland, Denmark, Netherlands, Australia.

3 to 6 gha (orange areas in map): New Zealand, Russia, Norway, Sweden, Western Europe (except Portugal), Saudi Arabia, Japan, Turkmenistan, Libya, Chile, Paraguay, Uruguay, Kuwait, United Arab Emirates.

0 to 3 gha (yellow areas in map): Mexico, Panama, Chile, Argentina, Brazil, Venezuela, Surinam, French Guiana, South Africa, Middle East and Eastern Europe, Kazakhstan, Chad, Burkina Faso, Central America, Colombia, Guyana, Ecuador, Peru, Bolivia, China, India, Central Asia, Mongolia, Indonesia, Central Africa.

Insufficient data (gray areas in map): Greenland, Iceland, Antarctica.

Earth has a total of 11.9 billion gha *of productive land. How many people could Earth support sustainably if the average ecological footprint were* 8 gha *per person (as in the United States)?*

*For suggested answer, see*Appendix A*.*

Our impact on the planet can also be assessed using currencies other than area, such as energy use. Average energy use differs greatly across different regions of the world **(Figure 53.25)**. A typical person in the United States, Canada, or Norway consumes roughly 30 times the energy that a person in central Africa does. Moreover, fossil fuels, such as oil, coal, and natural gas, are the source of 80% or more of the energy used in most developed nations. As Concept 56.4 discusses in more detail, this unsustainable reliance on fossil fuels is changing Earth's climate and increasing the amount of waste that humans produce. Ultimately, the combination of resource use per person and population density determines our global ecological footprint.

Figure 53.25

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The uneven electrification of the planet.
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A nighttime image of the planet reveals concentrated, erratic light patterns.

This composite image taken from space of Earth's surface at night illustrates the varied density of electric lights worldwide, one aspect of energy use by humans.

What factors will eventually limit the growth of the human population? Perhaps food will be the main limiting factor. Malnutrition and famine are common in some regions, but they result mainly from the unequal distribution of food rather than from inadequate production. So far, technological improvements in agriculture have allowed food supplies to keep up with global population growth. In contrast, the demands of many populations have already far exceeded the local and even regional supplies of one renewable resource—fresh water. More than 1 billion people do not have access to sufficient water to meet their basic sanitation needs. The human population may also be limited by the capacity of the environment to absorb its wastes. If so, then Earth's current human occupants could lower the planet's long-term carrying capacity for future generations.

Technology has substantially increased Earth's carrying capacity, but no population can grow indefinitely. After reading this chapter, you should realize that there is no single carrying capacity. How many people our planet can sustain depends on the quality of life each of us has and the distribution of wealth across people and nations, topics of great concern and political debate. We can decide whether zero population growth will be attained through social changes based on human choices or, instead, through increased mortality due to resource limitation, plagues, war, and environmental degradation.